Lung and Colon Cancer Histopathological Image Classification

Submitted by: Ishwari Gharat

1. Introduction

Histopathological image analysis involves examining tissue samples under a microscope. This project explores deep-learning image classification for five categories of lung and colon tissue. It compares a custom Convolutional Neural Network (CNN) with transfer learning using MobileNetV2. A Streamlit application was also developed to accept an image and display its predicted class and confidence score.

2. Dataset and Exploratory Analysis

The project uses the Lung and Colon Cancer Histopathological Images dataset published on Kaggle by Andrew (2020). It contains 25,000 RGB images across five balanced classes, with 5,000 images per class: colon_aca (colon adenocarcinoma), colon_n (normal colon tissue), lung_aca (lung adenocarcinoma), lung_n (normal lung tissue), and lung_scc (lung squamous cell carcinoma). The inspected images were 768 × 768 pixels with three RGB channels. Exploratory analysis included class distribution, representative images, image dimensions and a sample RGB intensity histogram.

3. Preprocessing

The dataset was divided using stratified sampling into training (17,500 images; 70%), validation (3,750; 15%) and test (3,750; 15%) subsets. Images were resized to 224 × 224 pixels, converted to RGB and scaled to [0,1]. Random flipping, rotation and zoom were used as training-time augmentation. The same image size and class ordering were used during evaluation and application inference.

4. Models and Results

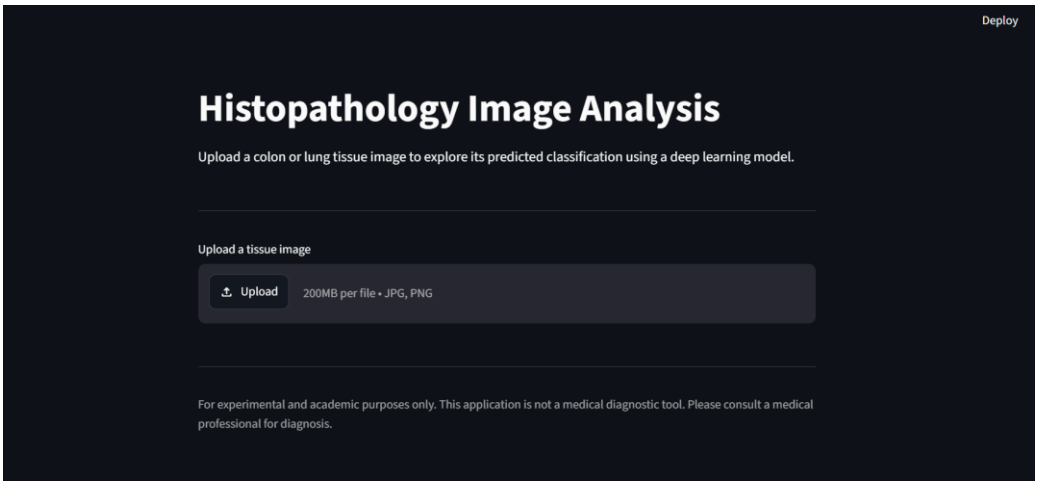
The custom CNN used convolutional and pooling layers to learn image features, followed by global average pooling and dense classification layers. MobileNetV2 used ImageNet-pretrained feature extraction with its base initially frozen and a new classification head trained for the five classes. MobileNetV2 is designed around inverted residuals and linear bottlenecks (Sandler et al., 2018).

Model	Test accuracy	Test loss
Custom CNN	98.51%	Not recorded
MobileNetV2	98.16%	0.0490

The custom CNN achieved 98.51% test accuracy, while MobileNetV2 achieved 98.16% with a test loss of 0.0490. MobileNetV2's classification report showed F1-scores of 99.60% for colon_aca, 99.67% for colon_n and 100% for lung_n. Lung_aca had lower recall (93.20%), indicating that some images in this class were assigned to another category. The confusion matrix and classification report are included in the results folder. The custom CNN test loss was not retained.

5. Application

A Streamlit interface was developed using the saved MobileNetV2 model. Users can upload a JPG, JPEG or PNG tissue image and request a prediction. The application displays the predicted class, a plain-language explanation and the model confidence score. A disclaimer explains that the application is for academic experimentation and is not a medical diagnostic tool.



Conclusion

The project completed image exploration, preprocessing, training and evaluation of two deep-learning models, and development of an image-upload application. The custom CNN achieved 98.51% recorded test accuracy and MobileNetV2 achieved 98.16%. Future work should use patient-level separation, independent external validation and expert pathology review before any clinical consideration.

